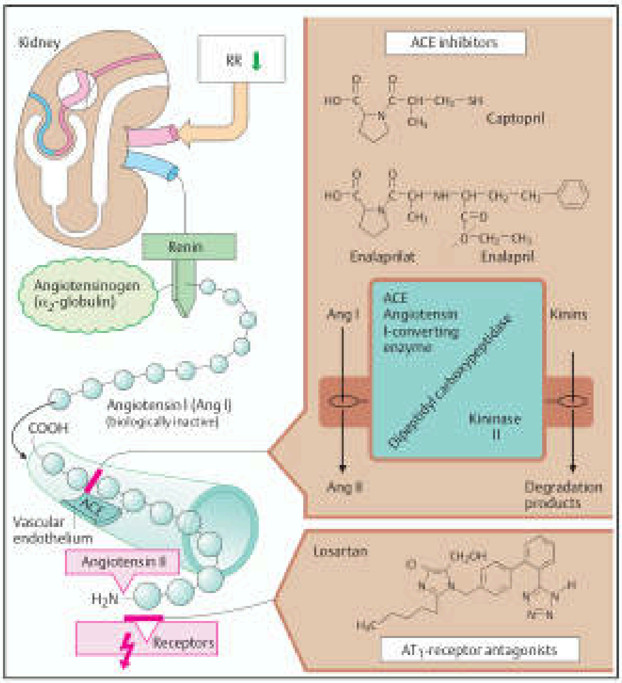


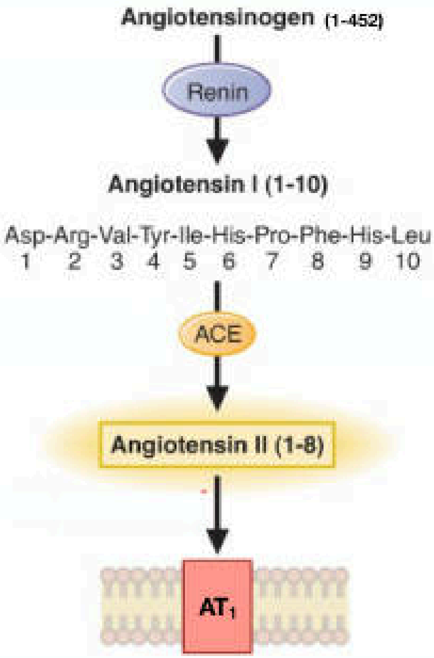
ACE Inhibitors, ARBs & DRIs

Krisada Sakchaisri, Ph.D.

วัตถุประสงค์

1. สามารถอธิบายกลไกการออกฤทธิ์และฤทธิ์ทางเภสัชวิทยาของยาที่ออกฤทธิ์ต่อระบบเรนิน-แองจิโอเทนซินกลุ่มต่างๆ ได้
2. สามารถบรรยายอาการไม่พึงประสงค์ อันเกี่ยวเนื่องจากฤทธิ์ทางเภสัชวิทยาของยาที่ออกฤทธิ์ต่อระบบเรนิน-แองจิโอเทนซินกลุ่มต่างๆ ได้
3. สามารถอธิบายบทบาทและประโยชน์ทางการแพทย์ของยาที่ออกฤทธิ์ต่อระบบเรนิน-แองจิโอเทนซินกลุ่มต่างๆ ได้

The Renin-Angiotensin System (RAS)



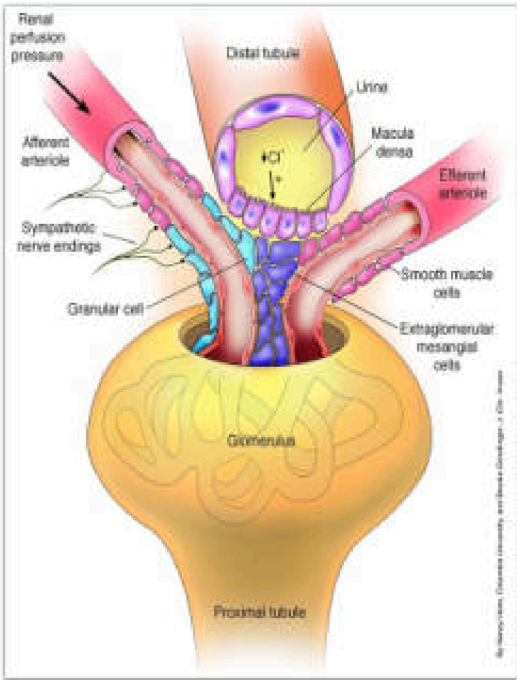
Angiotensinogen

- A glycoprotein precursor of angiotensin
- Synthesis
  - liver seems to be the major source
  - other tissues: brain, heart, adiposes, etc.
  - stimulated by inflammation, insulin, estrogens, glucocorticoids, thyroid hormone

Renin

- An aspartyl protease
- Synthesis and storage: juxtaglomerular (JG) apparatus
- The amount of renin in the bloodstream is a key **rate-limiting step** determining the level of Ang II and thus the activity of the RAS

Control of Renin Release



1. The renal baroreceptor pathway

: renal perfusion pressure

: ⬇ BP in the preglomerular vessels ➡ ⬆ renin secretion

2. The macula densa pathway

: sensing a reduction in Cl^- at the macular densa ➡ renin release

3. The β1 adrenergic receptor pathway

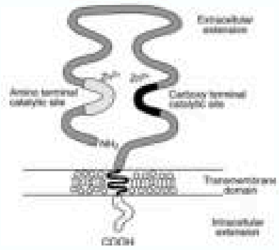
: Activation of β_1 receptors on JG cells ➡ ⬆cAMP and ⬆ renin secretion

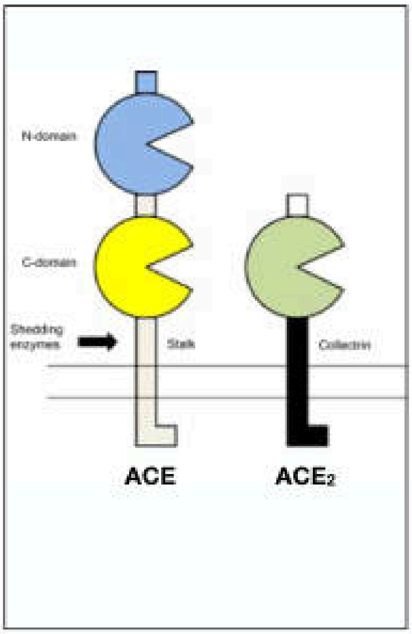
Feedback Regulation

: stimulate AT_1 receptor at JG cells inhibits renin release

: inhibition of renin release due to Ang II-dependent increase in BP

Angiotensin Converting Enzyme (ACE)





Angiotensin **C**onverting **E**nzyme

- ACE, kininase II
- A dicarboxypeptidase
- Expression
  : blood vessels, kidney, intestine, adrenal gland, liver, and uterus
  : the lung vascular endothelium contains the highest concentration of ACE
- Function
  : conversion of Ang I to Ang II
  : degradation of bradykinin (BK), a vasodilator peptide

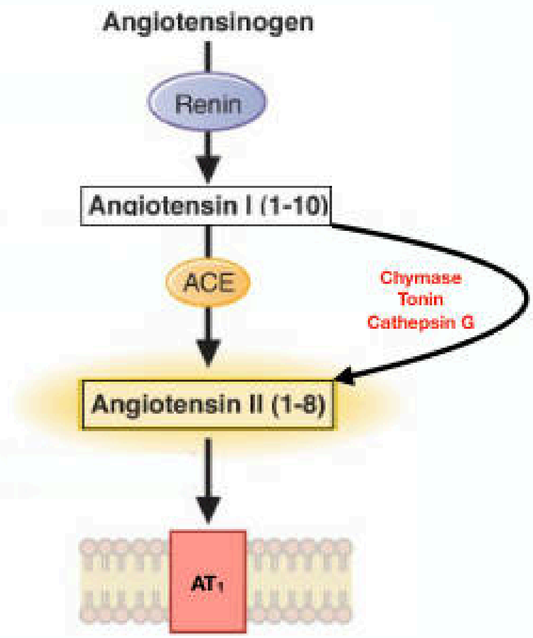
<u>Note</u>

- ACE is rather nonspecific: other targets including encephalin, neurotensin, and substance P
- Other functions: renal development, male fertility, hematopoiesis, and immune responses

Y. Takei, H. Ando, & K. Tsutsui (Eds): Handbook of Hormones.

Cardiovasc Drugs and Ther. 2002; 16: 149–160.

Alternative Pathways for Angiotensin II Biosynthesis



- Angiotensin II may be generated through ACE-independent pathways or “ACE escape”
  : Cathepsin G
  : Tonin
  : Chymase
- Chymase contributes to tissue conversion of Ang I and Ang(1–12) to Ang II, particularly in the heart and kidneys

Major Physiological Effects of Angiotensin II

Angiotensin II

	Altered peripheral resistance	Altered renal function	Altered cardiovascular structure
Mechanisms	• Direct **vasoconstriction** • Enhancement of peripheral noradrenergic neurotransmission • Increase sympathetic discharge (CNS) • Release catecholamine from adrenal medulla	• Release of **aldosterone** from adrenal cortex (↑ Na^+ reabsorption and ↑ K^+ excretion) • Altered renal hemodynamics : Direct renal vasoconstriction : ↑ noradrenergic neurotransmission in kidney : ↑ renal sympathetic tone	• **Hemodynamically effects** : ↑ after load : ↑ wall tension • **Nonhemodynamically mediated effects** : ↑ expression of proto-oncogenes : ↑ production of growth factors : ↑ synthesis of extracellular matrix proteins
Results	**Rapid pressor response**	**Slow pressor response**	**Vascular and cardiac hypertrophy & remodeling**

Angiotensin Receptors

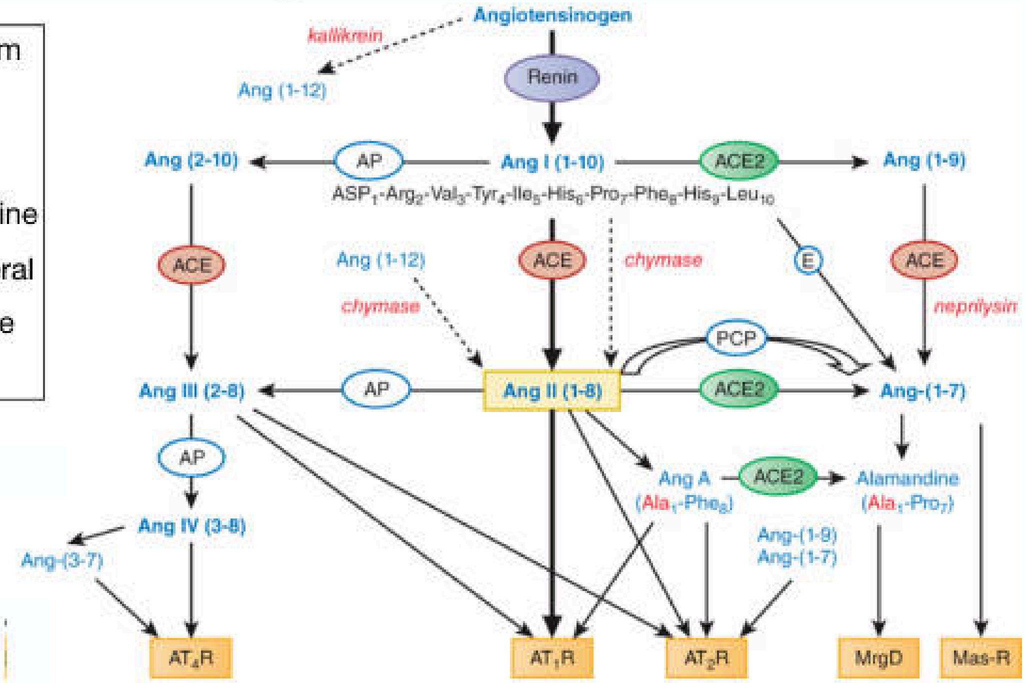
AT_1R	AT_2R
Hypertensive Effects	**Antihypertensive Effects**
Vasoconstriction	Vasodilation
Fibrosis	Antifibrotic
VSMC inflammation	Anti-inflammation
Oxidative stress	↓Oxidative stress
Cardiac hypertrophy	Antiproliferation
Disease	Protection

VSMC, Vascular Smooth Muscle Cell

Modified from Goodman and Gilman's the pharmacological basis of therapeutics

New Paradigms in The RAS

The RAS has expanded from being solely an endocrine system to include a paracrine, autocrine/intracrine hormonal system with several new components and active pathways



AP, aminopeptidase; E, endopeptidases; PCP, prolylcarboxylpeptidase

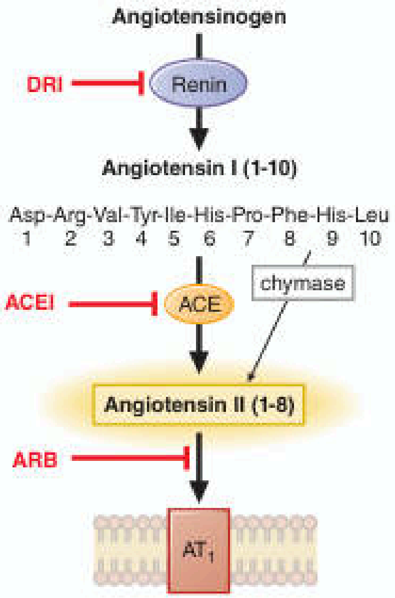
Modified from Goodman and Gilman's the pharmacological basis of therapeutics

Angiotensin Receptors

RECEPTOR	RAS PEPTIDE	EFFECT
AT_1	AngII, AngIII, AngA, Ang(1–12)	Vasoconstriction hypertrophy Fibrosis, nephropathy
AT_2	AngII, AngIII, Ang(1–7), Ang(1–9), AngA	Vasodilation, antihypertrophy, antifibrosis Natriuresis
Mas	Ang(1–7)	Vasodilation, antihypertrophy, antifibrosis Natriuresis
MrgD	Alamandine	Vasodilation, antihypertrophy, antifibrosis
AT_4	AngIV, Ang(3–7)	Neuroprotection Cognition Renal vasodilation Natriuresis
PRR (pro)renin receptor	Prorenin, renin	Hypertrophy, fibrosis Apoptosis

Goodman and Gilman's the pharmacological basis of therapeutics

Inhibitors of The Renin-Angiotensin System

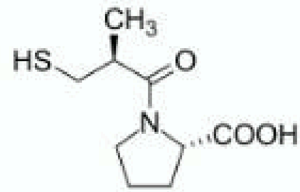


1. **ACE inhibitors**
2. **Angiotensin receptor blockers**
3. **Direct renin inhibitors**

All of these classes of agents will ***reduce the actions of Ang II*** *and* ***lower BP****, but each has different effects on the individual components of the RAS*

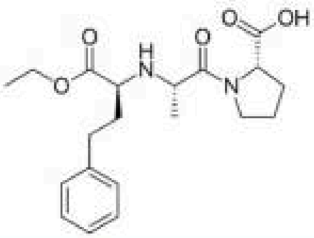
Angiotensin Converting Enzyme Inhibitors (ACE Inhibitors)

Classification of ACE Inhibitors



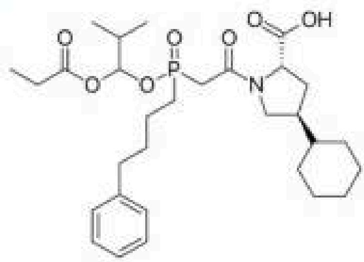
1. **Sulfhydryl-containing ACE inhibitors**

   : captopril



2. **Dicarboxyl-containing ACE inhibitors**

   : enalapril ,lisinopril, benazepril, quinapril, moexipril, ramipril, trandolapril, perindopril

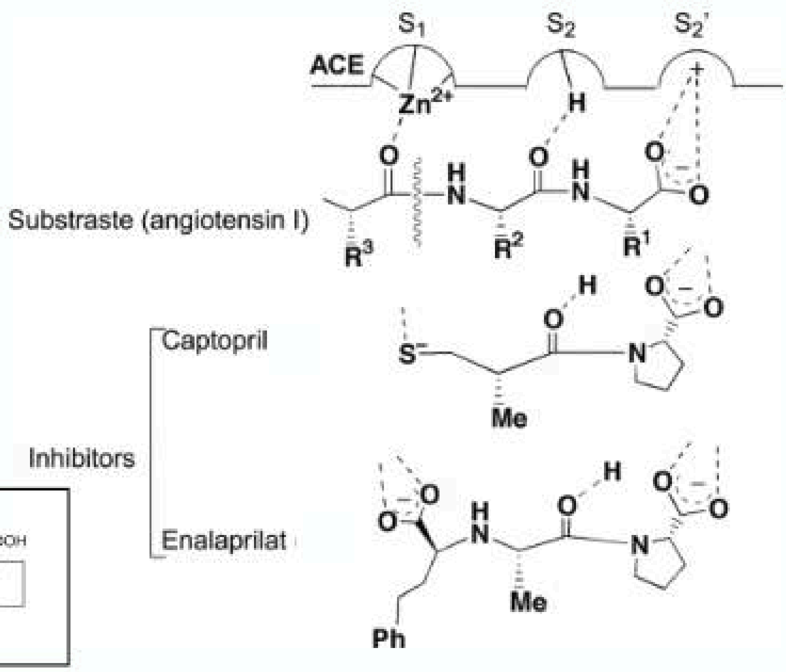


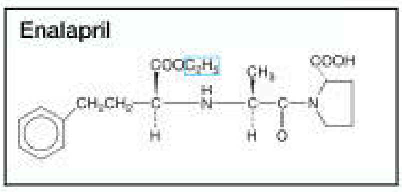
3. **Phosphorus-containing ACE inhibitors**

   : fosinopril

Many ACE inhibitors are ester-containing prodrugs that are 100–1000 times less potent but have better oral bioavailability than the active molecules

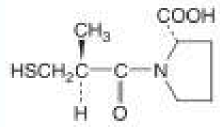
The Binding Mode of ACE Inhibitors



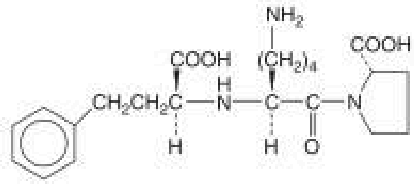


Examples of ACE Inhibitors

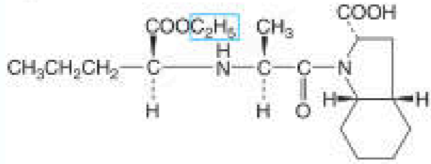
Captopril



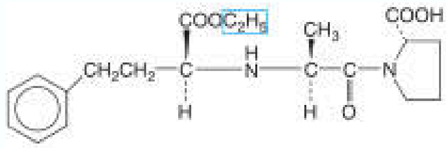
Lisinopril



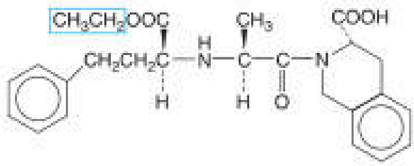
Perindopril



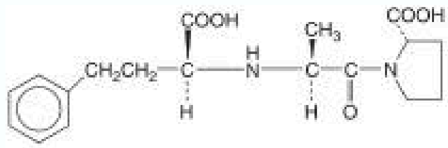
Enalapril



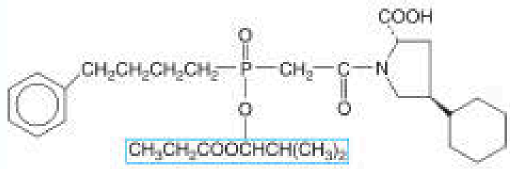
Quinapril



Enalaprilat



Fosinopril



Goodman and Gilman's the pharmacological basis of therapeutics

Pharmacokinetics of ACE Inhibitors

Name	Zinc Ligand	Prodrug	F	Onset of Action	Protein Binding	Half-Life	Route of Elimination
Captopril	Sulfhydryl	No	75-91%	0.25 h	25-30%	2 h	Kidney
Enalapril	Carboxyl	Yes	60%	1 h	<50%	11 h	Kidney
Lisinopril	Carboxyl	No	6-60%	1 h	25%	12 h	Kidney
Benazepril	Carboxyl	Yes	>37%	1 h	>95%	10-11 h	Kidney
Quinapril	Carboxyl	Yes	>60%	1 h	97%	2 h (initial), 25 h (terminal)	Kidney
Moexipril	Carboxyl	Yes	13%	1.5 h	50%	2-12 h	Kidney & Liver
Ramipril	Carboxyl	Yes	50-60%	1-2 h	56%	Triphasic: 2-4, 9-18, >50 h	Kidney
Perindopril	Carboxyl	Yes	75%	1-2 h	10-20%	Biphasic: 3-10, 30-120 h	Kidney & Liver
Trandolapril	Carboxyl	Yes	70%	4 h	80%	16-24 h	Kidney
Fosinopril	Phosphinyl	Yes	36%	1 h	95%	12 h	Kidney & Liver

Pharmacological Effects of ACE Inhibitors

- ACE inhibitors Inhibit the conversion of angiotensin I to angiotensin II

➡ lowering blood pressure

➡ enhancing natriuresis

Note

- ACE degrades vasodilator peptide bradykinin

  : ACE inhibitors ➝ having beneficial antihypertensive and protective effects

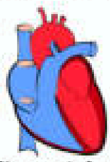
- ACE inhibitors ➝ ↑ renin release and ↑ angiotensin I (no negative feedbacks)

  : ➡ ↑Ang(1–9) and Ang(1–7) vasodilator peptides

Therapeutic Uses of ACE Inhibitors

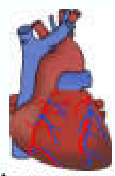
Hypertension

: ↓ systemic vascular resistance and mean, diastolic, and systolic BPs in various hypertensive states except high BP due to primary aldosteronism



Left ventricular systolic dysfunction

: ACE inhibitor delays the progression of heart failure, ↓ the incidence of sudden death and myocardial infarction, ↓ hospitalization, and improves quality of life



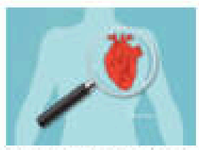
Acute myocardial infarction

Diabetes Mellitus and renal failure

: ACE inhibitors prevent or delay the progression of renal disease

Scleroderma renal crisis

: ACE inhibitors improve survival of these patients



Patients who are at risk of cardiovascular events

Adverse Effects of ACE Inhibitors

Initial dose hypotension

: A steep fall in BP may occur following the first dose of an ACE inhibitor

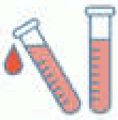
Dry cough

: 5–20%

: mediated by the accumulation in the lungs of bradykinin, substance P, or PGs

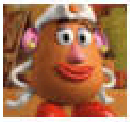
Acute renal failure

: in patients with bilateral renal artery stenosis, stenosis of the artery to a single remaining kidney, heart failure, or volume depletion



Hyperkalemia

: rarely encountered in patients with normal renal function



Angioedema

: 0.1-0.5%

: swelling in the nose, throat, mouth, glottis, larynx, lips, or tongue

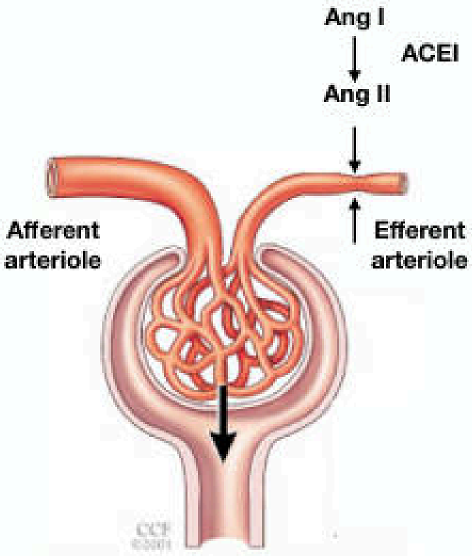
Fetopathic potential

: associated with fetal and neonatal injury, including fetal death to pregnant women during the 2nd and 3rd trimester

Skin rash

: maculopapular rash

Angiotensin II and Glomerular Filtration



- Ang II constricts the efferent arteriole to a greater extent than the afferent arteriole
  - ➟ Ang II increases GFR
- When these conditions occur in ACE inhibitor–treated patients
  - ➟ Ang II formation and effect are diminished
  - ➟ GFR may decrease
- Blockade of the RAS may cause acute renal failure in patients with bilateral renal artery stenosis or in patients with unilateral stenosis who have only a single kidney

Drug Interactions

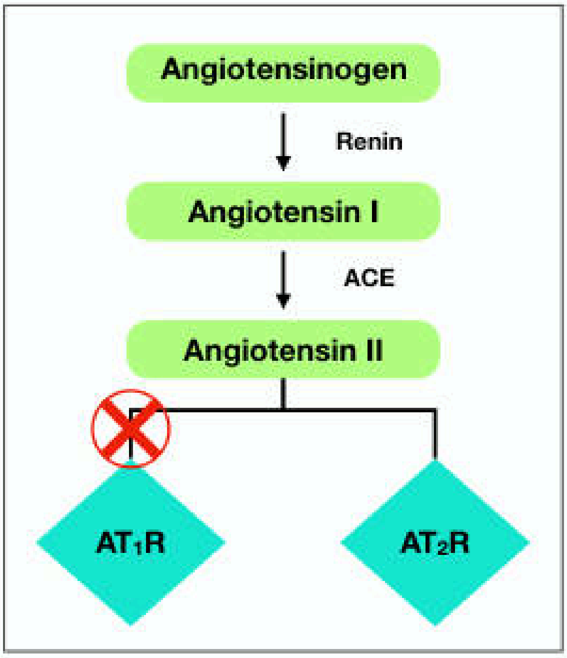
Agents	Description
NSAIDs	May reduce the antihypertensive response to ACE inhibitors
K^+-sparing diuretics	May exacerbate ACE inhibitor–induced hyperkalemia
K^+-supplements	May exacerbate ACE inhibitor–induced hyperkalemia
Digoxin	ACE inhibitors may increase plasma levels of digoxin
Lithium	ACE inhibitors may increase plasma levels of lithium
Capsaicin	May worsen ACE inhibitor–induced cough

Angiotensin Receptor Blockers (ARBs)

Rationale in The Development of ARBs

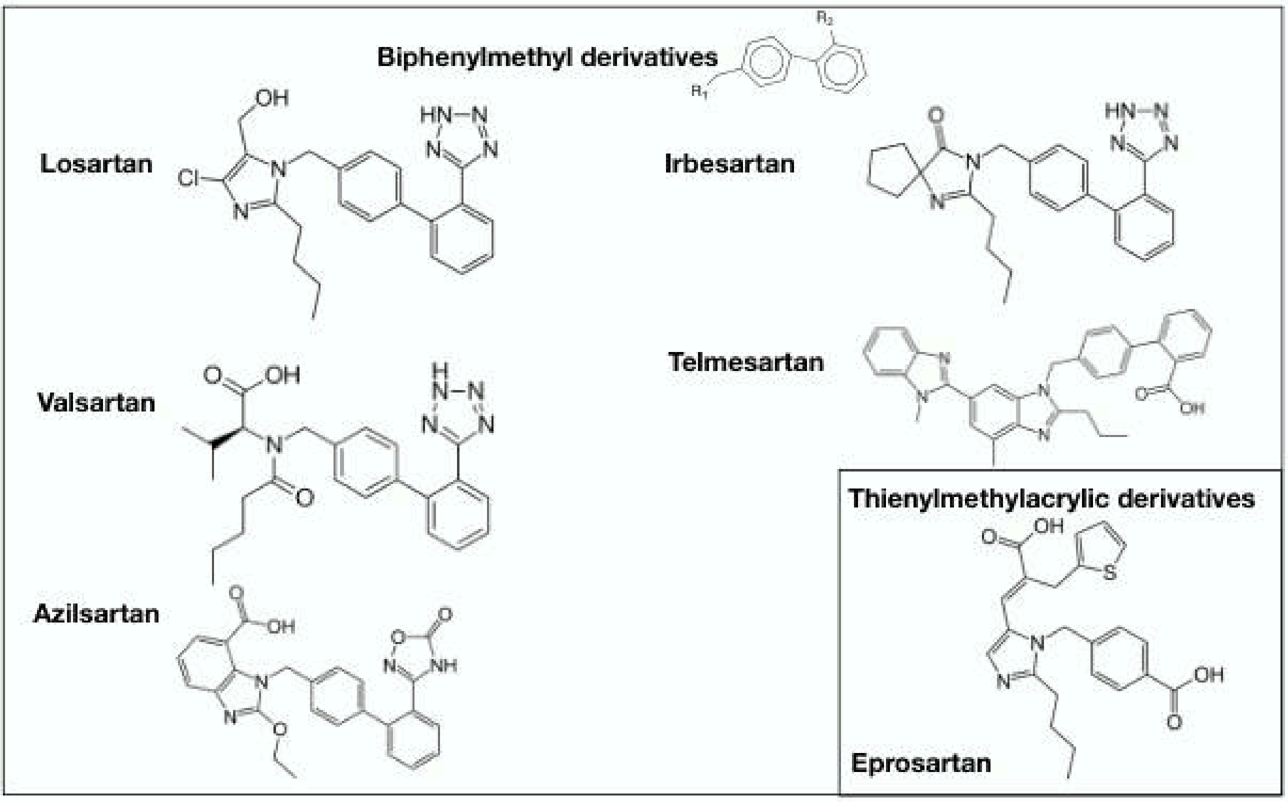
- Competitive inhibition of ACE results in a reactive increase in renin and Ang I levels, which may overcome the blockade effect
- ACE is a relatively non-specific enzyme that has substrates in addition to Ang I, including bradykinin and other tachykinins
  - ➠ inhibition of ACE may result in accumulation of these substrates
- Production of Ang II can occur through non-ACE pathways as well as through the primary ACE pathway
- Specific adverse effects are associated with ACE inhibitor effects on the enzyme
- ARBs may offer more complete Ang II inhibition by interacting selectively with the receptor site

Angiotensin II Receptor Blockers (ARBs)



- The ARBs' **mechanism of action**: selective inhibition of Ang II by competitive antagonism of the Ang II receptor (AT_1 receptor)
- The ARBs bind to the AT_1 receptor with high affinity and are more than 1,000-fold selective for the AT_1 receptor over the AT_2 receptor

ARBs



ARBs

Name	AT_1 Affinity vs AT_2	F	Protein Binding	Active Metabolite	Half-Life	Note
Losartan	1,000-fold	33%	99%	Yes	2 h	competitive antagonist of the thromboxane A2 receptor and attenuates platelet aggregation
Valsartan	30,000-fold	25%	95%	No	6 h	
Eprosartan	1,000-fold	15%	98%	No	5 h	
Irbesartan	>8,500-fold	70%	90%	NO	11-15 h	
Candesartan	10,000-fold	42%	99%	Yes	9 h	
Olmesartan	12,500-fold	26%	>99%	Yes	13	
Telmisartan	3,000-fold	43%	99.5%	No	24 h	
Azilsartan	~39,000-fold	60%	99%	No	11 h	

Pharmacological Effects of ARBs

ARBs inhibit most of the biological effects of angiotensin II

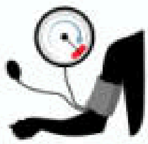
➡ ↓ *contraction of vascular smooth muscle*

➡ ↓ *rapid & slow pressor responses*

- ➡ ↓ *aldosterone secretion*
- ➡ ↓ *noradrenergic neurotransmission*

➡ ↓ *cardiac muscle hypertrophy and hyperplasia*

Therapeutic Uses of ARBs

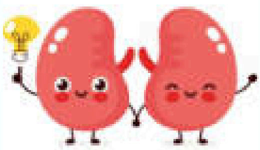


Hypertension

: All ARBs are approved for the treatment of hypertension
: The efficacy of ARBs in lowering blood pressure is comparable with that of ACE inhibitors, with a favorable adverse-effect profile

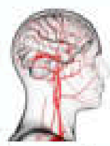
Heart failure

: Valsartan and candesartan are approved for heart failure and to reduce cardiovascular mortality in clinically stable patients with left ventricular failure or left ventricular dysfunction following myocardial infarction



Renoprotective

: ARBs are renoprotective in diabetic patients
: Irbesartan and losartan are approved for diabetic nephropathy



Stroke prophylaxis

: losartan is approved for stroke prophylaxis in patients with hypertension and left ventricular hypertrophy (LVH

Adverse Effects of ARBs

Initial dose hypotension

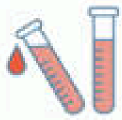
Teratogenic potential

: should be discontinued in pregnancy

Acute renal failure

: In patients whose renal function is highly dependent on the RAS (e.g., renal artery stenosis), ARBs can cause oliguria, progressive azotemia, or acute renal failure

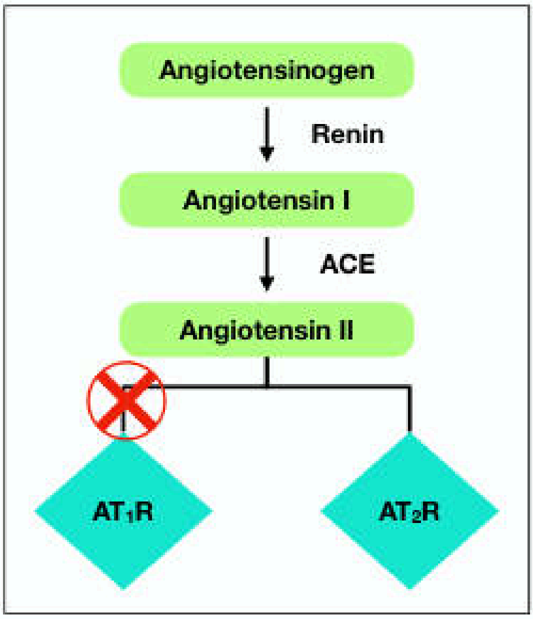
Angioedema



Hyperkalemia

- The ARBs are generally well tolerated
- The incidence of angioedema and cough with ARBs is less than that with ACE inhibitors

ACE Inhibitors VS ARBs



- ARBs ↓ activation of AT_1 receptors > ACE inhibitors

  ➡ *ACE inhibitors do not inhibit ACE-independent Ang II-producing pathways*

- ARBs permit activation of AT_2 receptors

  ➡ *ARBs block AT_1 receptors, this increased level of AngII is available to activate AT_2 receptors*

- ACE inhibitors increase the levels of a number of ACE substrates, including bradykinin

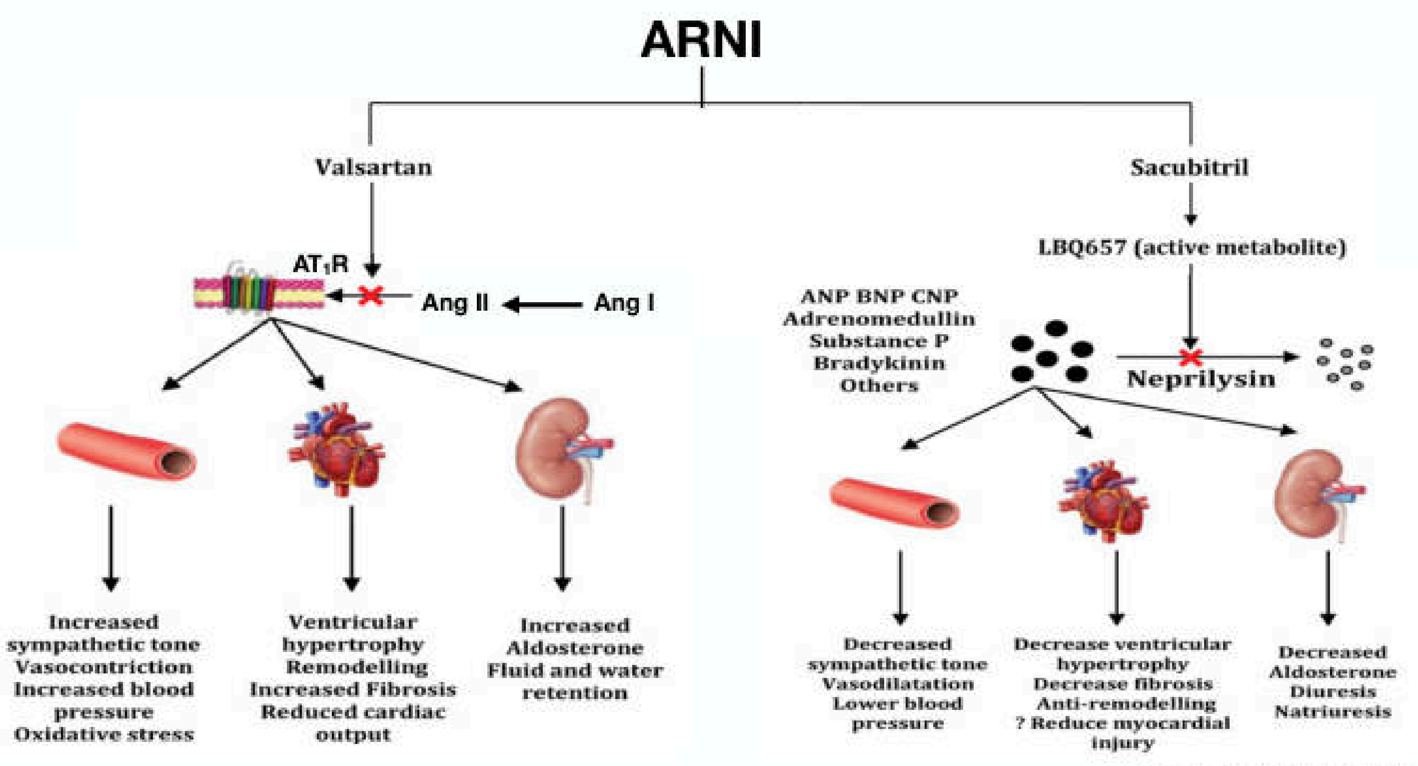
Angiotensin Receptor–Neprilysin Inhibitors (ARNIs)

- Valsartan/sacubitril (LCZ696, Novartis Pharmaceuticals) is a first-in-class angiotensin II-receptor neprilysin inhibitor
- It is approved for treatment of heart failure with reduced ejection fraction
- It is contraindicated in conjunction with an ACE inhibitor or in patients with a history of angioedema during ACE inhibitor or ARB use
- It should not be used in conjunction with an ARB or ACE inhibitor

Neprilysin

- Neprilysin is a zinc-dependent metallopetidase that catalyses the degradation of various peptides including ANP, BNP, and bradykinin, as well as contributes to the breakdown of angiotensin II
- Its inhibition results in natriuretic, vasodilatatory, and anti-proliferative effects

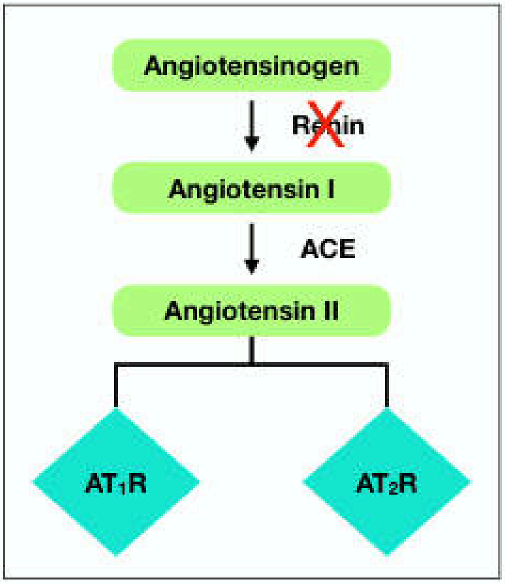
ANP, atrial natureuretic peptide; BNP, B-type natriuretic peptide



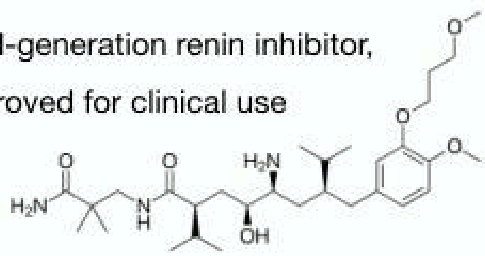
Eur Heart J (2015) 36, 1967–1973.

Direct Renin Inhibitors (DRIs)

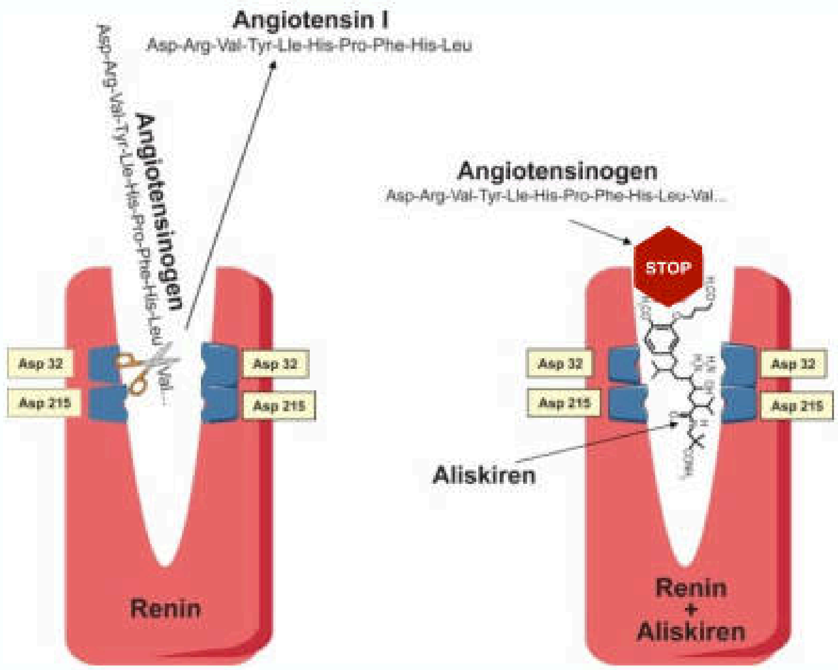
Direct Renin Inhibitors (DRIs)



- DRIs inhibit the cleavage of Ang I from angiotensinogen by renin
- First-generation renin inhibitors (enalkiren, zankiren, CGP38560A, and remikiren) were effective in reducing Ang II levels, but none of them made it past clinical trials due to their low potency, poor bioavailability, and short $t_{1/2}$
- **Aliskiren**, a second-generation renin inhibitor, is the only DRI approved for clinical use



Pharmacological Effects of Aliskiren



- A low-molecular-weight nonpeptide and a potent competitive inhibitor of renin ➡ reducing the production of Ang II

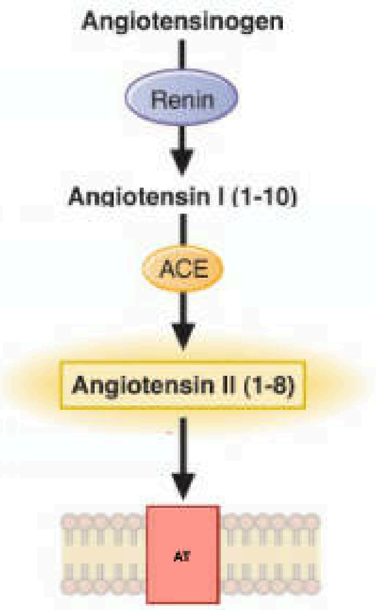
Vasc Health Risk Manag. 2008 Oct; 4(5): 971–981.

Adverse Effects of Aliskiren

- Mild GI symptoms: e.g., abdominal pain, dyspepsia
- Headache
- Nasopharyngitis
- Dizziness
- Fatigue
- Upper respiratory tract infection
- Back pain
- Angiodema
- Cough (much less common than with ACE inhibitors)

Aliskiren is well tolerated, and adverse events are mild or comparable to placebo with no gender difference

Comparison of ACE Inhibitors, ARBs & DRIs



	DRIs	ACE Inhibitors	ARBs
Plasma renin	↑	↑	↑
Plasma renin activity	↓	↑	↑
Angiotensin I	↓	↑	↑
Angiotensin II	↓	↓	↑
Angiotensin (1-7)	↓	↑	↑
Bradykinin	-	↑	-
AT_1R	Not stimulated	Not stimulated	Blocked
AT_2R	Not stimulated	Not stimulated	Stimulated

Drugs	Therapeutic Uses	Clinical Pharmacology and Tips
Angiotensin-Converting Enzyme Inhibitors : Inhibit the conversion of Angiotensin I to Angiotensin II		
Captopril **Lisinopril** **Enalapril** **Benazepril** **Quinapril** **Ramipril** **Moexipril**	• Inhibit Ang II production, thus lowering arteriolar resistance • Hypertension • Acute myocardial infarction • Congestive heart failure • Diabetic nephropathy • Scleroderma renal crisis	• Antihypertensive effects potentiated by inhibition of ACE-catalyzed breakdown of bradykinin • Antihypertensive effects potentiated by increase in Ang(1–7) levels and activation of Ang(1–7)/Mas receptor pathway • Increase PRC and PRA • Adverse effects include cough in 5%–20% of patients, angioedema, hypotension, hyperkalemia, skin rash, neutropenia, anemia, fetopathic syndrome • Contraindicated in patients with renal artery stenosis and should be used with caution in patients with impaired renal function or hypovolemia • Should be stopped during pregnancy

PRC, plasma renin concentration; PRA, plasma renin activity

Drugs	Therapeutic Uses	Clinical Pharmacology and Tips
Angiotensin Receptor Blockers : Block AT1 receptors		
Losartan **Valsartan** **Eprosartan** **Irbesartan** **Candesartan** **Olmesartan** **Telmisartan** **Azilsartan**	• Block the vasoconstrictor and profibrotic effects of Ang II by inhibiting AT1 receptors while permitting vasodilation through activation of AT2 receptors • Hypertension • Congestive heart failure • Diabetic nephropathy	• Antihypertensive effects potentiated by activation of the AT2 receptors • Antihypertensive effects potentiated by ACE2-dependent conversion of AngII to Ang(1–7) and activation of vasodilation via Ang(1–7)/Mas receptor pathway • Increase PRC and PRA • Adverse effects include hyperkalemia and hypotension • Contraindicated in patients with renal insufficiency • Should be stopped during pregnancy

PRC, plasma renin concentration; PRA, plasma renin activity

Drugs	Therapeutic Uses	Clinical Pharmacology and Tips
Direct Renin Inhibitors : Inhibit renin and thus the conversion of angiotensinogen to Angiotensin I		
Aliskiren	• Decrease Angiotensin I and Angiotensin II levels • Hypertension	• Therapeutic value unclear; no evidence for superiority over ACEIs or ARBs • Increase PRC but decrease PRA • Contraindicated in diabetic nephropathy, pregnancy or renal insufficiency

PRC, plasma renin concentration; PRA, plasma renin activity